

Experiment 5: Constructor Overloading – Employee Data

1. Question

Write a Java program to demonstrate constructor overloading using an Employee class. Create multiple constructors with different numbers of parameters and display the employee details.

2. Steps

1. Start the program.
2. Create an Employee class.
3. Declare private data members: name, id, designation, salary, and promotionstatus.
4. Create a default constructor with no parameters.
5. Initialize default values in the default constructor.
6. Create a two-parameter constructor with name and id.
7. Create a four-parameter constructor with name, id, designation, and salary.
8. Create a five-parameter constructor with all employee details.
9. Define the show() method to display employee details.
10. Create four employee objects using different constructors.
11. Call show() for each employee object.
12. Display the employee details.
13. Stop the program.

3. Algorithm

1. Start.
2. Define the Employee class.
3. Declare employee attributes.
4. Define a default constructor and assign default values.
5. Define a constructor with two parameters.
6. Define a constructor with four parameters.
7. Define a constructor with five parameters.
8. Define the show() method.
9. Create e1 using the default constructor.
10. Create e2 using the two-parameter constructor.
11. Create e3 using the four-parameter constructor.
12. Create e4 using the five-parameter constructor.
13. Display details of all employees using show().
14. Stop.

4. UML Diagram

```

+-----+
|           Employee           |
+-----+
| - name : String              |
| - id : int                   |
| - designation : String       |
| - salary : double            |
| - promotionstatus : boolean  |
+-----+
| + Employee()                 |
| + Employee(name, id)         |
| + Employee(name, id,         |
|   designation, salary)       |
| + Employee(name, id,         |
|   designation, salary,       |
|   promotionstatus)           |
| + show() : void              |
+-----+

```

Constructor Overloading

```

Employee()
|
+---- Employee(String name, int id)
|
+---- Employee(String name, int id,
|           String designation, double salary)
|
+---- Employee(String name, int id,
|           String designation, double salary,
|           boolean promotionstatus)

```

The same class has multiple constructors with different parameter lists. This is called constructor overloading.

5. Code

```

package packagecodejava;

class Employee {

    private String name;
    private int id;
    private String designation;
    private double salary;
    private boolean promotionstatus;

    // Default constructor
    public Employee() {
        this.name = "not assigned";
        this.id = 0;
        this.designation = "trainer";
        this.salary = 0.0;
        this.promotionstatus = false;
    }

    // Constructor with 2 parameters
    public Employee(String name, int id) {
        this.name = name;
        this.id = id;
        this.designation = "trainer";
        this.salary = 0.0;
        this.promotionstatus = false;

        System.out.println("--- 2 parameters passing constructor");
    }

    // Constructor with 4 parameters
    public Employee(String name, int id, String designation, double salary) {
        this.name = name;

```

```

        this.id = id;
        this.designation = "trainer";
        this.salary = 0.0;
        this.promotionstatus = false;

        System.out.println("4paramters passing constructor");
    }

    // Constructor with 5 parameters
    public Employee(String name, int id, String designation,
                    double salary, boolean promotionstatus) {

        this.name = name;
        this.id = id;
        this.designation = "trainer";
        this.salary = 0.0;
        this.promotionstatus = false;

        System.out.println("all details passing constructor");
    }

    public void show() {

        System.out.println("--- employee details ---");
        System.out.println("name : " + name);
        System.out.println("id : " + id);
        System.out.println("designation : " + designation);
        System.out.println("salary : " + salary);
        System.out.println("promotionstatus : " + promotionstatus);
    }
}

public class Week05 {

    public static void main(String[] args) {

        Employee e1 = new Employee();
        e1.show();

        Employee e2 = new Employee("hemanth", 244);
        e2.show();

        Employee e3 = new Employee(
            "hemanth", 244, "software engineer", 70.000);
        e3.show();

        Employee e4 = new Employee(
            "hemanth", 244, "software engineer", 75.000, false);
        e4.show();
    }
}

```

6. Correct Output

```

--- employee details ---
name : not assigned
id :0
designation : trainer
salary :0.0
promotionstatus : false
--- 2 parameters passing constructor
--- employee details ---
name : hemanth
id :244
designation : trainer
salary :0.0
promotionstatus : false
4paramters passing constructor
--- employee details ---
name : hemanth

```

eclipse-workspace - packagecodejava/src/packagecodejava/Week05.java - Eclipse IDE

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Tasks Console X

<terminated> Week05 [Java Application] C:\Users\Subba Reddy\Downloads\eclipse-jee-2026-06-R-win32

```
--- employee details ---
name : not assigned
id :0
designation : trainer
salary :0.0
promotionstatus : false
--- 2 parameters passing constructor
--- employee details ---
name : hemanth
id :244
designation : trainer
salary :0.0
promotionstatus : false
4paramters passing constructor
--- employee details ---
name : hemanth
id :244
designation : trainer
salary :0.0
promotionstatus : false
all details passing constructor
```

46 public void show(){

```
****
Tasks Console X
<terminated> Week05 [Java Application] C:\Users\Subba Reddy\Downloads\eclipse-jee-2026-06-R-w
--- employee details ---
name : hemanth
id :244
designation : trainer
salary :0.0
promotionstatus : false
4paramters passing constructor
--- employee details ---
name : hemanth
id :244
designation : trainer
salary :0.0
promotionstatus : false
all details passing constructor
--- employee details ---
name : hemanth
id :244
designation : trainer
salary :0.0
promotionstatus : false
```